Task 11: Recommend and Defend the Operating-System Plan

Operating Systems (CSA04) — Assignment 1 (CO1)

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Marks: 4 Marks (Bloom's: K5 - Evaluate)
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> ***Task 11 Description:*** *Recommend the complete OS master plan for the hospital. Specify OS type, structure, services, protection, and I/O methods for each environment. Address plan limitations, risks, and scaling across 3 hospitals. Evaluate the alternative of virtual machines on two large servers.*

1. CONSOLIDATED OPERATING-SYSTEM ARCHITECTURE PLAN

Environment	OS Type & Structure	I/O Architecture	Protection & Isolation
500 Bedside Monitors	Hard RTOS (QNX / FreeRTOS) with Microkernel	Interrupt-driven, Memory-Mapped I/O, priority inheritance	Hardware MPU/MMU boundaries; user-space driver sandboxing
120 Nurses' Tablets	Handheld Mobile OS (Android AOSP) with Modular kernel	Asynchronous duty-cycled power management	SELinux enforcing, per-application UID sandboxing, full disk encryption
80 Doctors' Desktops	Enterprise Desktop (Linux/Windows) with Hybrid/Modular	Standard VFS buffer cache with device driver frameworks	Hardware ring protection (Ring 3 vs 0), Smart-card PAM auth
Central Database	Clustered HA OS (RHEL + Pacemaker) with Monolithic/LKM	Direct I/O (`O_DIRECT` , `io_uring`), NVMe DMA	Kernel memory namespaces, role-based access control, encrypted volumes
MRI Imaging Server	Multiprocessor NUMA SMP Linux with Monolithic structure	Direct Memory Access (DMA) over PCIe buses	Process memory isolation, high-priority CUDA thread pinning
Partner Link	Distributed Network OS (Linux / K8s Gateway)	Non-blocking network event polling (`epoll`)	Zero-Trust mutual TLS 1.3, hardware cryptoprocessors (TPM/HSM)

2. EVIDENCE-BASED DEFENSE & MULTI-HOSPITAL SCALING

- **Microkernels for Monitors:** While microkernel IPC incurs a ~3µs overhead, this consumes only 0.03% of the 10ms deadline budget, providing driver crash isolation that prevents fatal monitoring failures.
- **Monolithic HA Cluster for Database:** Retaining drivers and the page cache in Ring 0 supports > 5,000 TPS without IPC context-switching penalties. Automatic quorum fail-over prevents single points of failure.
- **Three-Hospital Federation:** As the system scales to 1,500 monitors and 360 tablets, distributed network gateways handle cross-facility traffic asynchronously, preserving local ward determinism.

3. ALTERNATIVE CONCEPT: CONSOLIDATING EVERYTHING ONTO TWO LARGE VM HOSTS

- **Loss of Real-Time Determinism:** Running bedside monitors inside hypervisor virtual machines (e.g., VMware ESXi, KVM) introduces "noisy neighbor" interference. Heavy MRI reconstruction workloads cause CPU scheduling jitter, vCPU preemption, and memory page overcommit delays, pushing alarm latencies past 50–100ms and violating patient safety.
- **Single Point of Failure Concentration:** Hosting 500 monitors across two physical servers concentrates operational risk. A hypervisor crash or hardware failure on Host 1 instantly takes down 250 clinical monitors simultaneously.